# 16

# The Singular Value Decomposition

**Historical Note**

No theorem in this book has been discovered so many times. Beltrami found the singular value decomposition in 1873 and Jordan, independently, in 1874, both working with bilinear forms on square real matrices. Sylvester found it again in 1889 and thought it new. Autonne extended it to complex matrices in 1915, and Eckart and Young gave the general rectangular statement in 1936, along with the approximation theorem that now carries their names, though Erhard Schmidt, of Chapter 12, had proved an infinite-dimensional version of that in 1907. The name comes from the theory of integral equations, where the reciprocals of these numbers were the *singular* values of a parameter at which an equation failed to have a unique solution. A practical algorithm arrived only in 1965, from Golub and Kahan, and after that the decomposition became what it is now: probably the single most useful factorisation in applied mathematics.

Chapter 14 gave a complete answer for matrices that are square and self-adjoint, and no answer at all for anything else. But most matrices in practice are rectangular, and a rectangular matrix has no eigenvalues, since $Ax = \lambda x$ cannot even be written when $Ax$ and $x$ live in different spaces. This chapter shows what survives. The answer is that one orthonormal basis is not enough, and two are exactly enough.

## 16.1 Two bases instead of one

The route in is to make a symmetric matrix out of $A$ by force.

**Proposition 16.1** ($A^tA$ and $AA^t$)**.** *Let $A$ be a real $m \times n$ matrix.*

*(a) $A^tA$ is symmetric and positive semidefinite, so its eigenvalues are real and nonnegative.*

*(b) $A^tA$ and $AA^t$ have the same nonzero eigenvalues, with the same multiplicities.*

***Proof.*** (a) $(A^tA)^t = A^tA$, and $\langle A^tAx, x\rangle = \langle Ax, Ax\rangle = \|Ax\|^2 \geq 0$; now apply Theorem 14.14.

(b) Suppose $A^tAv = \lambda v$ with $\lambda \neq 0$ and $v \neq \mathbf{0}$. Then $\|Av\|^2 = \langle A^tAv, v\rangle = \lambda\,\|v\|^2 \neq 0$, so $Av \neq \mathbf{0}$, and

$$AA^t(Av) = A\big(A^tAv\big) = \lambda\, Av,$$

so $Av$ is an eigenvector of $AA^t$ for the same $\lambda$. The map $v \mapsto Av$ is injective on that eigenspace, since $Av = \mathbf{0}$ would force $\lambda v = \mathbf{0}$, so the multiplicity for $AA^t$ is at least that for $A^tA$. Exchanging the roles of $A$ and $A^t$ gives the reverse inequality. □

**Definition 16.2** (Singular values)**.** *The* singular values *of a real $m \times n$ matrix $A$ are the nonnegative square roots*

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$$

*of the eigenvalues of $A^tA$, listed with multiplicity.*

**Key Idea**

Singular values exist for every matrix of every shape, are always real, and are always nonnegative. Compare eigenvalues, which need a square matrix, may be complex, and may fail to supply a basis at all. The price is that we shall need two orthonormal bases rather than one, and the whole content of the next theorem is that two suffice.

## 16.2 The decomposition

**Theorem 16.3** (Singular value decomposition)**.** *Every real $m \times n$ matrix $A$ can be written*

$$A = U\Sigma V^t$$

*where $U$ is $m \times m$ orthogonal, $V$ is $n \times n$ orthogonal, and $\Sigma$ is the $m \times n$ matrix with $\sigma_1, \dots, \sigma_n$ down its main diagonal and zeros elsewhere. Writing $v_i$ and $u_i$ for the columns,*

$$Av_i = \sigma_i u_i \quad (i \leq r), \qquad Av_i = \mathbf{0} \quad (i > r),$$

*where $r$ is the number of nonzero singular values. The same statement holds over $\mathbb{C}$ with $U, V$ unitary and $^t$ replaced by $^*$.*

Before the proof, it is worth knowing what the statement looks like. Figure 16.1

draws a $2 \times 2$ matrix carrying the unit circle to an ellipse, and names the four objects in the theorem on it: two perpendicular directions in the source, the two lengths they are stretched by, and the two perpendicular directions they land on. Every matrix is a rotation, a stretch along perpendicular axes, and another rotation.

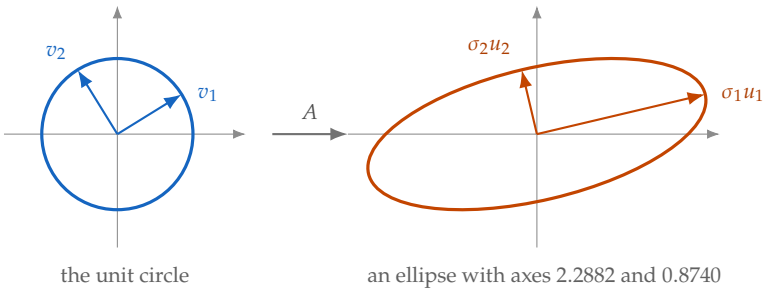



**Figure 16.1.** Every matrix is a rotation, a stretch along perpendicular axes, and another rotation. Here $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ carries the unit circle to an ellipse. The singular value decomposition names the four objects in the picture: $v_1, v_2$ are the perpendicular directions in the source that $A$ sends to perpendicular directions, $u_1, u_2$ are their images after normalising, and $\sigma_1 = 2.288246$, $\sigma_2 = 0.874032$ are the stretch factors. Nothing about $A$ suggested perpendicularity, and the theorem is that such a pair always exists. The product $\sigma_1\sigma_2 = 2$ equals $|\det A|$, which is why the ellipse has exactly twice the area of the circle, and the ratio $\sigma_1/\sigma_2 = 2.618$ is the condition number.

***Proof.*** By Proposition 16.1(a) and Theorem 14.7, $\mathbb{R}^n$ has an orthonormal basis $v_1, \dots, v_n$ with $A^tAv_i = \sigma_i^2 v_i$, ordered so that $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Let $r$ be the number of nonzero $\sigma_i$ and put

$$u_i = \frac{Av_i}{\sigma_i} \qquad \text{for } i \leq r.$$

These are orthonormal: for $i, j \leq r$,

$$\langle u_i, u_j \rangle = \frac{\langle Av_i, Av_j \rangle}{\sigma_i\sigma_j} = \frac{\langle A^tAv_i, v_j \rangle}{\sigma_i\sigma_j} = \frac{\sigma_i^2 \langle v_i, v_j \rangle}{\sigma_i\sigma_j} = \delta_{ij}.$$

Extend $u_1, \dots, u_r$ to an orthonormal basis $u_1, \dots, u_m$ of $\mathbb{R}^m$, which Theorem 12.10 allows. For $i > r$ we have $\|Av_i\|^2 = \langle A^tAv_i, v_i \rangle = \sigma_i^2 = 0$, so $Av_i = \mathbf{0}$. Hence $Av_i = \sigma_i u_i$ for every $i$, reading $\sigma_i = 0$ when $i > r$, which is exactly $AV = U\Sigma$. Multiplying on the right by $V^t$ finishes. □

**Corollary 16.4** (The four subspaces, all at once)**.** *With notation as above,* $r = \operatorname{rank} A$ *and*

$$C(A) = \operatorname{span}\{u_1, \dots, u_r\}, \qquad N(A^t) = \operatorname{span}\{u_{r+1}, \dots, u_m\},$$

$$R(A) = \operatorname{span}\{v_1, \dots, v_r\}, \qquad N(A) = \operatorname{span}\{v_{r+1}, \dots, v_n\},$$

*and each pair is an orthonormal basis of its subspace.*

***Proof.*** Every $Ax$ is $\sum_i \sigma_i \langle x, v_i \rangle u_i$, a combination of $u_1, \dots, u_r$, and each $u_i$ with $i \le r$ is $Av_i/\sigma_i \in C(A)$; so $C(A) = \operatorname{span}\{u_1, \dots, u_r\}$ and $\operatorname{rank} A = r$. The vectors $v_{r+1}, \dots, v_n$ lie in $N(A)$ and there are $n - r$ of them, which is $\dim N(A)$ by Corollary 6.17, so they are a basis of it; the remaining two identities follow from Lemma 13.8 and Theorem 13.2, since the two families are orthonormal complements within each space. □

**Key Idea**

That single corollary supplies orthonormal bases for all four subspaces of Theorem 13.12 at one stroke, which no amount of elimination will do. Elimination gives bases; the SVD gives *orthonormal* bases, ordered by how much of the matrix's action each direction carries. It also settles a debt from Chapter 6: by Proposition 16.1(b) the matrices $A$ and $A^t$ have the same nonzero singular values, so by the corollary they have the same rank. That is the equality of row rank and column rank once more, and this time the reason is that both count the same list of numbers rather than the same pivots.

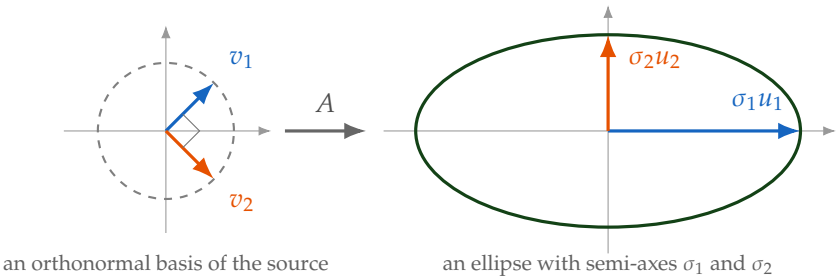



**Figure 16.2.** Theorem 16.3 for $A = \left(\begin{smallmatrix} 2 & 2 \\ 1 & -1 \end{smallmatrix}\right)$, whose singular values are $\sigma_1 = 2\sqrt{2}$ and $\sigma_2 = \sqrt{2}$. The map carries the two perpendicular directions $v_1, v_2$ to the two perpendicular directions $u_1, u_2$, stretching each by its own singular value, so the unit circle goes to an ellipse. What makes this possible for *every* matrix is that the input frame and the output frame are allowed to differ: here $v_1$ points at 45° while $u_1$ is the first standard basis vector.

**Example 16.5** (An SVD, in full). *Let* $A = \left(\begin{smallmatrix} 2 & 2 \\ 1 & -1 \end{smallmatrix}\right)$. *Then*

$$A^t A = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix},$$

*whose eigenvalues are* 8 *and* 2, *with eigenvectors* $(1,1)$ *and* $(1,-1)$; *these are perpendicular, as Proposition 14.4(b) guarantees. So* $\sigma_1 = 2\sqrt{2}$ *and* $\sigma_2 = \sqrt{2}$, *and*

$$v_1 = \tfrac{1}{\sqrt{2}}(1,1), \qquad v_2 = \tfrac{1}{\sqrt{2}}(1,-1).$$

*Now* $Av_1 = \frac{1}{\sqrt{2}}(4,0)$, *of norm* $2\sqrt{2} = \sigma_1$, *so* $u_1 = (1,0)$; *and* $Av_2 = \frac{1}{\sqrt{2}}(0,2)$, *of norm* $\sqrt{2} = \sigma_2$, *so* $u_2 = (0,1)$. *Hence*

$$U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}, \qquad V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

*Check by multiplying out:* $\Sigma V^t = \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} 2\sqrt{2} & 2\sqrt{2} \\ \sqrt{2} & -\sqrt{2} \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2 & 2 \\ 1 & -1 \end{smallmatrix}\right) = A$, *and* $U = \mathbf{1}$ *here. Two further checks worth making every time:* $\sigma_1\sigma_2 = 4 = |\det A|$, *and* $\sigma_1^2 + \sigma_2^2 = 10 = 4 + 4 + 1 + 1$, *the sum of the squares of the entries.*

## 16.3 Outer products, and the thin form

**Proposition 16.6** (Sum of rank-one pieces). *With notation as in Theorem 16.3,*

$$A = \sigma_1 u_1 v_1^t + \sigma_2 u_2 v_2^t + \cdots + \sigma_r u_r v_r^t,$$

*a sum of* $r$ *matrices each of rank one. Equivalently* $A = U_r \Sigma_r V_r^t$, *the* thin *form, where* $U_r$ *and* $V_r$ *keep only the first* $r$ *columns and* $\Sigma_r$ *is the* $r \times r$ *diagonal matrix of nonzero singular values.*

***Proof.*** Apply both sides to an arbitrary $x$. Since $v_i^t x = \langle x, v_i \rangle$, the right side gives

$$\sum_{i \le r} \sigma_i \, u_i \, (v_i^t x) = \sum_{i \le r} \sigma_i \, \langle x, v_i \rangle \, u_i,$$

and that is $Ax$, by Theorem 16.3 together with Theorem 12.9. Each $u_i v_i^t$ has rank one because every one of its columns is a multiple of $u_i$. □

**Example 16.7.** *For the matrix of Example 16.5,*

$$A = 2\sqrt{2}\cdot\begin{pmatrix}1\\0\end{pmatrix}\tfrac{1}{\sqrt{2}}(1,1) + \sqrt{2}\cdot\begin{pmatrix}0\\1\end{pmatrix}\tfrac{1}{\sqrt{2}}(1,-1) = \begin{pmatrix}2 & 2\\0 & 0\end{pmatrix} + \begin{pmatrix}0 & 0\\1 & -1\end{pmatrix},$$

*and the two pieces do indeed add to $A$. The first carries the larger part of the matrix, in a sense the next section makes precise.*

Cutting the sum short is what makes the decomposition useful in practice. Figure 16.3 shows the arithmetic of that truncation: keeping the first $k$ terms replaces $mn$ stored entries by $k(m+n+1)$.

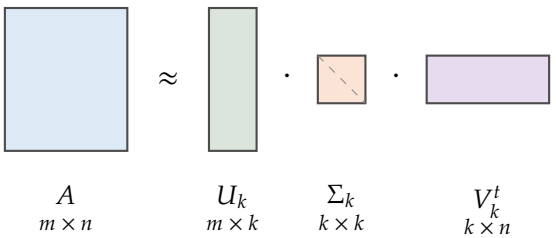



**Figure 16.3.** The rank-$k$ truncation of Proposition 16.6. Keeping only the first $k$ singular values and their vectors replaces $mn$ entries by $k(m+n+1)$, which for small $k$ is an enormous saving; and Theorem 16.12 says no other matrix of rank $k$ does better. The dashed line in $\Sigma_k$ marks its diagonal, the only place it is nonzero.

## 16.4 Norms and conditioning

**Definition 16.8** (Two matrix norms)**.** *For a real $m\times n$ matrix $A$, the* spectral norm *and the* Frobenius norm *are*

$$\|A\|_2 = \max_{\|x\|=1}\|Ax\|, \qquad \|A\|_F = \Big(\sum_{i,j} a_{ij}^2\Big)^{1/2}.$$

*The second is the norm coming from the Frobenius inner product of Example 12.2(iii).*

**Proposition 16.9** (Both norms are singular values)**.** $\|A\|_2 = \sigma_1$ *and* $\|A\|_F = \big(\sigma_1^2 + \cdots + \sigma_n^2\big)^{1/2}$.

***Proof.*** For $\|x\| = 1$ write $x = \sum_i c_i v_i$ with $\sum_i c_i^2 = 1$. Then $Ax = \sum_i \sigma_i c_i u_i$ and, the $u_i$ being orthonormal, $\|Ax\|^2 = \sum_i \sigma_i^2 c_i^2 \le \sigma_1^2$, with equality at $x = v_1$. For the Frobenius norm, $\|A\|_F^2 = \operatorname{trace}(A^t A)$ by Example 12.2(iii), and the trace is the sum of the eigenvalues by Theorem 9.12, which for $A^t A$ are the $\sigma_i^2$. $\square$

**Definition 16.10** (Condition number)**.** *For an invertible square $A$, the* condition number *is $\kappa(A) = \sigma_1/\sigma_n$.*

**Proposition 16.11.** *For $A$ invertible, $\kappa(A^t A) = \kappa(A)^2$.*

***Proof.*** $A^t A$ is symmetric positive definite with eigenvalues $\sigma_1^2, \dots, \sigma_n^2$. For such a matrix $M$ the singular values are the eigenvalues, because $M^t M = M^2$ has eigenvalues the squares. So the singular values of $A^t A$ are the $\sigma_i^2$ and $\kappa(A^t A) = \sigma_1^2/\sigma_n^2 = \kappa(A)^2$. $\square$

**Key Idea**

That closes the account opened in Chapter 13. The Caution there asserted that squaring a matrix squares its condition number, and that is why the normal equations must not be formed; here is the proof, and it is one line once the singular values are available. The condition number also has a direct meaning: it is the factor by which $A$ can amplify a relative error, since the worst input direction is stretched by $\sigma_1$ and the best by $\sigma_n$.

## 16.5 Best low-rank approximation

**Theorem 16.12** (Eckart–Young)**.** *For $k < r$ put $A_k = \sum_{i \le k} \sigma_i u_i v_i^t$. Then for every matrix $B$ of rank at most $k$,*

$$\|A - B\|_2 \;\ge\; \|A - A_k\|_2 = \sigma_{k+1},$$

*and likewise $\|A - B\|_F \ge \|A - A_k\|_F = \left(\sigma_{k+1}^2 + \cdots + \sigma_r^2\right)^{1/2}$. So $A_k$ is a closest rank-$k$ matrix to $A$ in both norms.*

***Proof.*** That $\|A - A_k\|_2 = \sigma_{k+1}$ is Proposition 16.9 applied to $A - A_k = \sum_{i>k} \sigma_i u_i v_i^t$, whose singular values are $\sigma_{k+1}, \dots, \sigma_r$; the Frobenius statement is the same computation.

For the lower bound, let $\operatorname{rank} B \le k$. Then $\dim N(B) \ge n - k$ by Corollary 6.17, while $\operatorname{span}\{v_1, \dots, v_{k+1}\}$ has dimension $k+1$. Since $(n-k) + (k+1) > n$, Exercise 6.31 gives a

nonzero vector $w$ in both; scale it to $\|w\| = 1$. Now $Bw = \mathbf{0}$, so

$$\|A - B\|_2^2 \;\geq\; \|(A-B)w\|^2 = \|Aw\|^2 .$$

Writing $w = \sum_{i \leq k+1} c_i v_i$ with $\sum c_i^2 = 1$, we get $\|Aw\|^2 = \sum_{i \leq k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2$, because $\sigma_i \geq \sigma_{k+1}$ for every $i \leq k+1$. Hence $\|A - B\|_2 \geq \sigma_{k+1}$.

The Frobenius lower bound is a separate argument, because a single vector $w$ no longer carries the whole norm. Let rank $B \leq k$ and let $P$ be the orthogonal projection onto $C(B)$, so $P$ has rank $s \leq k$. Every column $b_j$ of $B$ lies in $C(B)$, so Theorem 13.4 applied one column at a time gives

$$\|A - B\|_F^2 = \sum_j \|a_j - b_j\|^2 \;\geq\; \sum_j \|a_j - Pa_j\|^2 = \|A - PA\|_F^2 = \|A\|_F^2 - \|PA\|_F^2 ,$$

the last step by Pythagoras in each column. Now let $q_1, \dots, q_s$ be an orthonormal basis of $C(B)$. Since $P$ is symmetric and idempotent, $\|Pa\|^2 = \langle Pa, a \rangle$, so

$$\|PA\|_F^2 = \sum_j \langle PAe_j, Ae_j \rangle = \operatorname{trace}\!\left(P\,AA^t\right) = \sum_{i \leq s} q_i^t (AA^t) q_i .$$

By Proposition 16.6, $AA^t = \sum_{j \leq r} \sigma_j^2 u_j u_j^t$, so with $t_j := \sum_{i \leq s} \langle q_i, u_j \rangle^2$ this reads $\|PA\|_F^2 = \sum_{j \leq r} \sigma_j^2 t_j$. Bessel's inequality bounds these weights twice over. It follows from Theorem 12.9 by extending an orthonormal family to an orthonormal basis and discarding the extra terms, which are nonnegative: $t_j \leq \|u_j\|^2 = 1$ for each $j$, and $\sum_{j \leq r} t_j \leq \sum_{i \leq s} \|q_i\|^2 = s \leq k$. A sum $\sum_j \sigma_j^2 t_j$ with weights in $[0,1]$ totalling at most $k$ is largest when the weight sits on the $k$ largest $\sigma_j^2$, so $\|PA\|_F^2 \leq \sigma_1^2 + \cdots + \sigma_k^2$. Therefore

$$\|A - B\|_F^2 \;\geq\; \|A\|_F^2 - (\sigma_1^2 + \cdots + \sigma_k^2) = \sigma_{k+1}^2 + \cdots + \sigma_r^2 ,$$

using $\|A\|_F^2 = \sum_j \sigma_j^2$ from Proposition 16.9. □

**Example 16.13** (Best rank one). *For $A = \left(\begin{smallmatrix} 2 & 2 \\ 1 & -1 \end{smallmatrix}\right)$ of Example 16.5, the best rank-one approximation is the first outer product from Example 16.7,*

$$A_1 = \begin{pmatrix} 2 & 2 \\ 0 & 0 \end{pmatrix}, \qquad A - A_1 = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix},$$

*and indeed $\|A - A_1\|_2 = \sqrt{2} = \sigma_2$. No other rank-one matrix comes closer, in either norm. Notice that $A_1$ is not obtained by rounding or deleting entries of $A$: it is the projection of*

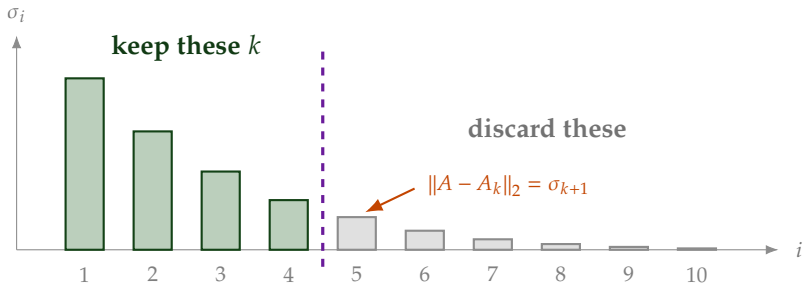



**Figure 16.4.** Theorem 16.12 in practice. The singular values of a matrix arising from real data typically decay quickly, so truncating after $k$ terms discards little; and the error incurred is not merely bounded by the first discarded singular value, it is exactly equal to it. That is what makes the theorem usable: the spectrum tells you in advance what any given truncation will cost.

*A onto the rank-one matrices, and which entries survive is decided by the singular vectors rather than by size.*

## 16.6 The pseudoinverse

**Definition 16.14** (Pseudoinverse)**.** *With $A = U\Sigma V^t$, let $\Sigma^+$ be the $n \times m$ matrix with $1/\sigma_i$ in position $(i,i)$ for $i \le r$ and zeros elsewhere. The* pseudoinverse *of $A$ is $A^+ = V\Sigma^+U^t$.*

**Theorem 16.15** (Minimum-norm least squares)**.** *For any $A$ and $b$, the vector $\hat{x} = A^+b$ is a least squares solution of $Ax = b$, and it is the one of smallest norm. If $A$ is invertible then $A^+ = A^{-1}$.*

***Proof.*** By Theorem 16.3, $Ax = \sum_{i \le r} \sigma_i \langle x, v_i \rangle u_i$, so $Ax$ is determined by the numbers $\langle x, v_i \rangle$ for $i \le r$ and nothing else. By Corollary 16.4 the projection of $b$ onto $C(A)$ is $\sum_{i \le r} \langle b, u_i \rangle u_i$, so by Theorem 13.9 the least squares solutions are exactly the $x$ with

$$\langle x, v_i \rangle = \frac{\langle b, u_i \rangle}{\sigma_i} \qquad (i \le r),$$

the coordinates $\langle x, v_i \rangle$ for $i > r$ being unconstrained. By Theorem 12.9 the norm of $x$ is smallest when those free coordinates are zero, which gives $x = \sum_{i \le r} \sigma_i^{-1} \langle b, u_i \rangle v_i$, and that is exactly $V\Sigma^+U^tb$. If $A$ is invertible then $r = n = m$ and $\Sigma^+ = \Sigma^{-1}$, so $A^+ = V\Sigma^{-1}U^t = (U\Sigma V^t)^{-1}$. □

**Key Idea**

This finishes Chapter 13. When $A$ has dependent columns the normal equations have infinitely many solutions, and Exercise 13.22 asked which one to prefer; the answer is the one lying in $R(A)$, with no component in the null space, and the pseudoinverse is precisely the machine that returns it. Note also what $A^+$ is not: it is not a two-sided inverse. $AA^+$ is the projection onto $C(A)$ and $A^+A$ the projection onto $R(A)$, and they are the identity only when $A$ is invertible.

**Insight**

Almost every large-scale use of linear algebra in machine learning is this chapter. Principal component analysis is the SVD of the centred data matrix, and taking it directly rather than forming a covariance matrix is the same discipline as Proposition 13.11: never square what you can factor. Compressing a matrix of embeddings, or an image, is Theorem 16.12 with $k$ chosen from a picture like Figure 16.4. Low-rank adaptation of a large model trains an update $\Delta W = BC$ with $B$ and $C$ thin, which is a bet that the useful update has small rank and that Eckart–Young therefore loses little. And the reason a model's weight matrices are compared through their singular values, as Chapter 7 claimed, is Proposition 16.9: those numbers describe the map and not the coordinates it happens to be written in.

**Open Problem 16.16** (Low rank approximation in the wrong norm)**.** *The truncated singular value decomposition solves* $\min_{\operatorname{rank} B \le k} \|A - B\|$ *for the spectral norm, the Frobenius norm, and in fact every unitarily invariant norm. Change the norm to the entrywise* $\ell_1$ *norm,* $\sum_{ij} |A_{ij} - B_{ij}|$*, and the problem changes character completely.* Is there a polynomial-time algorithm achieving a constant-factor approximation for entrywise $\ell_1$ low rank approximation?

What is known. *The problem is NP-hard even for* $k = 1$*. Song, Woodruff and Zhong gave a polynomial-time algorithm with approximation ratio* $\operatorname{poly}(k \log n)$ *[575], and no constant-factor algorithm is known for any* $k \ge 1$*; nor is constant-factor hardness known. The same question for the entrywise* $\ell_\infty$ *norm is wider open still.*

Why the norm matters so much. *The proof that the truncated singular value decomposition is optimal rests on the fact that a unitarily invariant norm depends only on the singular values, so the problem can be transported into a problem about the numbers* $\sigma_i$ *and solved there by an interlacing argument, which is Chapter 16's central mechanism. The entrywise* $\ell_1$ *norm is not unitarily invariant: rotating* $A$ *changes it. So there is no*

*transport, no reduction to scalars, and no reason for the answer to have anything to do with the singular values.* The singular value decomposition solves one norm's problem exquisitely and tells you nothing about the next norm along.

Where a reader might start. *Take* $A = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ *and find the best rank-one approximation in the Frobenius norm, where it is* $\frac{1}{3}J$ *for the all-ones-like structure given by the top singular vector, and then in the entrywise* $\ell_1$ *norm, where a rank-one matrix supported on a single row does better. The two answers are different objects, not just different numbers, and that difference is the whole problem.*

## Exercises

**Exercise 16.17.** *Find the singular values and a full SVD of each matrix.*

*(a)* $\left(\begin{smallmatrix} 3 & 0 \\ 0 & -2 \end{smallmatrix}\right)$

*(b)* $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$

*(c)* $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$

**Exercise 16.18.** *Let $A$ be real symmetric with eigenvalues $\lambda_i$. Show that the singular values of $A$ are the $|\lambda_i|$, and that the SVD may be taken with $U = V$ exactly when $A$ is positive semidefinite. What is the SVD of the negative identity $-\mathbf{1}_n$?*

**Exercise 16.19.** *Prove that for square $A$, $|\det A| = \sigma_1 \sigma_2 \cdots \sigma_n$. Interpret this using Theorem 8.19 and Figure 16.2.*

**Exercise 16.20.** *For $A = \left(\begin{smallmatrix} 2 & 2 \\ 1 & -1 \end{smallmatrix}\right)$, compute $\|A\|_2$, $\|A\|_F$ and $\kappa(A)$. Then verify by hand that $\|A\|_2 \le \|A\|_F \le \sqrt{r}\,\|A\|_2$, and say when each inequality is an equality.*

**Exercise 16.21.** *Let $A = uv^{t}$ with $u, v$ nonzero. Show that* $\operatorname{rank} A = 1$*, that $\sigma_1 = \|u\|\,\|v\|$ and that all other singular values are zero. Write down the SVD explicitly.*

**Exercise 16.22.** *For $A = \left(\begin{smallmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{smallmatrix}\right)$ and $b = (1, 2, 4)$, compute $A^{+}$ from the SVD and check*

*that $A^{+}b$ agrees with the minimum-norm solution you found in Exercise 13.22.*

**Exercise 16.23.** *Show that $AA^{+}$ is the orthogonal projection onto $C(A)$ and $A^{+}A$ the orthogonal projection onto $R(A)$. Deduce that $A^{+}AA^{+} = A^{+}$ and $AA^{+}A = A$.*

**Exercise 16.24.** *Let $A$ have singular values $10, 6, 3, 1, 1, 1, 1$. What is the smallest $k$ for which the rank-$k$ truncation satisfies $\|A - A_k\|_F \leq \frac{1}{10}\|A\|_F$? Answer the same question for the spectral norm and explain why the two answers differ.*

**Exercise 16.25.** *Give a $2 \times 2$ matrix with $\kappa(A) = 10^6$ whose entries are all of size about $1$. Then compute $\kappa(A^{t}A)$ and say, in terms of double precision arithmetic, why forming $A^{t}A$ would be a mistake.*

## References and further reading

Stewart's survey of the early history of the decomposition is the standard account of who found what and when, and it is a good corrective to the impression that mathematics proceeds tidily. Trefethen and Bau [607] open their book with the SVD rather than saving it for the end, on the grounds that it is the right first fact about matrices, and their treatment of conditioning and of the geometry in Figure 16.2 is the one we have followed. Axler [37] derives the decomposition from the polar decomposition, which makes the relationship to the spectral theorem clearest. Banica [51], our baseline, treats singular values as the natural generalisation of eigenvalues for the operator-theoretic material of its later parts, and obtains the equality of row rank and column rank as a corollary, which is the illuminating proof that Chapter 6 promised and did not give. For Eckart–Young in its many forms, including the versions for other norms due to Mirsky, and for the uses in data analysis, the literature is enormous; Austin [35] is a gentle entry point and treats image compression concretely.